

Due on Tuesday, February 20th.

If pictures are needed/relevant, please provide them with your solutions.

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1. Potential from Field.

The electric field in some region is $\mathbf{E} = E_0 \hat{i}$. We will consider the electric potential difference, $V_B - V_A$, between two points A and B on the x - y plane, with coordinates $(0, 3L)$ and $(4L, 0)$ respectively. Ignore the z -coordinate for this problem. Here E_0 and L are positive constants.

Recall: $V_B - V_A = - \int \mathbf{E} \cdot d\mathbf{l}$, with the line integral proceeding along any curve starting at point A and ending at point B .

- (a) [1 pt.] Sketch the situation, showing the x and y axis, the two points A and B , and the direction of the electric field.
- (b) [1 pt.] Consider the straight line pointing from A to B . Any element $d\mathbf{l}$ on this line is at the same angle with the direction of the electric field. Calling this angle θ , we obtain $\mathbf{E} \cdot d\mathbf{l} = E_0 |d\mathbf{l}| \cos \theta$. Calculate $\cos \theta$ based on your drawing.
- (c) [4 pts.] Calculate $V_B - V_A = - \int \mathbf{E} \cdot d\mathbf{l}$, integrating along the straight line pointing from A to B . The value of $\cos \theta$, calculated above, might be helpful.
- (d) [4 pts.] Consider now a path consisting of two segments: a line from A to point C at $(4L, 3L)$, and a line from point C to point B . Draw the situation. Only one of these (AC or CB) segments contribute to the line integral for $V_B - V_A$. Which one? Calculate $V_B - V_A$ by integrating along this segment.

2. Thin ring of charge.

A circular ring of radius R lies in the x - y plane with its center at the origin. The ring carries a uniformly distributed positive charge Q .

- (a) [10 pts.] Find the electric field due to the charged ring at the point $(0, 0, z_0)$ on the z -axis.
- (b) [5 pts.] Sketch plots of each component of the electric field as a function of z_0 .

3. Infinite plane of uniform electric charge.

The x - y plane ($z = 0$) is covered uniformly with electric charge, with surface charge density σ (i.e., any area A of the plane carries charge σA). Assume σ to be positive. We will calculate the electric field at some point P away from the plane. Without loss of generality, we can take the point to be $(0, 0, z_0)$, i.e., a point on the z -axis.

- (a) [**3 pts.**] By symmetry, what should be the direction of the electric field $\mathbf{E}$ at the point P ? What should be the direction of the electric field at any other point with positive z -coordinate? What about points with negative z -coordinate?

- (b) [**1+1 pts.**] Consider an infinitely long strip of the charged plane along the x -direction, running from $x = -\infty$ to $x = +\infty$, parallel to the x -axis. Let the strip be infinitesimally thin, located between $y = u$ and $y = u + du$.

Draw this situation in two ways, showing (1) A top view of the xy plane, and (2) a three-dimensional view also showing the z -axis and the point P .

- (c) [**4 pts.**] What is the linear charge density of the strip?
- (d) [**7 pts.**] Recall from the previous week: an infinite thin rod of charge, if carrying linear charge density λ , produces a field

$$\frac{1}{2\pi\epsilon_0} \frac{\lambda}{r}$$

at a point which is at radial distance r from the rod. In the present geometry, what is the magnitude and direction of the element of field ($d\mathbf{E}$) produced at P by our infinitesimal strip of charge?

Hint: a picture of the yz plane, showing the strip position and the point P , is useful, and is expected.

- (e) [**3 pts.**] Find the z -component of $d\mathbf{E}$.
- (f) [**5 pts.**] We now want to use the principle of superposition to obtain the net field at P . This means summing over all infinitesimal strips to cover the complete plane of charge, i.e., integrating your expression from $u = -\infty$ to $u = \infty$. It might help to remember that

$$\int \frac{dv}{v^2 + 1} = \tan^{-1} v \quad \text{and} \quad \lim_{v \rightarrow \pm\infty} \tan^{-1} v = \pm \frac{\pi}{2}$$

Note that you only need to deal with the z -component; the other components are zero by symmetry.

- (g) [**1 pt.**] What is the dependence of the net field on z_0 ?